Module 3: Sigma-Algebras and Measurable Spaces

Measurable structures, generators, Borel sets, and the information viewpoint

1. Sigma-Algebras

1.1 Motivation: which subsets can be measured?

A length or probability assignment needs a domain of sets. One might try $\mu : \mathcal{P}(\mathbb{R}) \to [0, \infty]$ with interval lengths $\mu([a, b]) = b - a$, translation invariance $\mu(A + t) = \mu(A)$, and *countable additivity* on pairwise disjoint sets. These natural requirements cannot hold simultaneously on *all* subsets of $\mathbb{R}$ (in the usual set-theoretic setting).

Vitali counterexample (proof sketch). On $[0, 1]$, write $x \sim y \iff x - y \in \mathbb{Q}$, and choose exactly one representative from each equivalence class; call the resulting set V. This selection uses the axiom of choice. For $q \in \mathbb{Q} \cap [-1, 1]$, let $V_q = V + q$. By the representative property, the sets V_q are pairwise disjoint, and

$$[0, 1] \subseteq U := \bigcup_{q \in \mathbb{Q} \cap [-1,1]} V_q \subseteq [-1, 2].$$

Were every subset measurable, translation invariance and countable additivity would imply, with $a = \mu(V)$,

$$1 \le \mu(U) \le 3, \qquad \mu(U) = \sum_{q \in \mathbb{Q} \cap [-1,1]} \mu(V_q) = \sum_{q \in \mathbb{Q} \cap [-1,1]} a = \begin{cases} 0, & a = 0, \\ +\infty, & a > 0. \end{cases}$$

Both possibilities contradict $1 \le \mu(U) \le 3$. Thus a length-like measure must restrict its domain; the Vitali set cannot be Lebesgue measurable. The restriction should still allow complements and countable set operations.

1.2 Definition and elementary properties

A *sigma-algebra* on Ω is a family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ such that

$$\boxed{\begin{aligned} &\Omega \in \mathcal{F}, \\ &A \in \mathcal{F} \Longrightarrow A^c \in \mathcal{F}, \\ &A_n \in \mathcal{F}\ (n \ge 1) \Longrightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}. \end{aligned}}$$

Its members are called *measurable sets*. By complements and De Morgan's laws, $\varnothing \in \mathcal{F}$ and $\bigcap_n A_n \in \mathcal{F}$ whenever $A_n \in \mathcal{F}$; finite unions, intersections, and differences are also measurable. Consequently, the lim sup and lim inf of any sequence of measurable sets are measurable.

Examples are the *trivial* sigma-algebra $\{\varnothing, \Omega\}$ and the *discrete* sigma-algebra $\mathcal{P}(\Omega)$. For $\Omega = \{1, 2, 3, 4\}$,

$$\mathcal{F} = \{\varnothing, \{1, 2\}, \{3, 4\}, \Omega\}$$

is a nontrivial example: it distinguishes the two blocks but not the individual points within either block.

1.3 Countable, not arbitrary, additivity

The *sigma-algebra axioms* govern which unions stay in the domain; *countable additivity* is a separate axiom for a measure μ defined on that domain:

$$\boxed{A_i \cap A_j = \varnothing\ (i \ne j) \quad \Longrightarrow \quad \mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).}$$

Arbitrary (possibly uncountable) additivity is *not* required. For instance, $[0, 1] = \bigcup_{x \in [0,1]} \{x\}$, and each singleton has Lebesgue length zero, yet $\lambda([0, 1]) = 1$. Countable additivity permits countable limiting operations without asserting an uncountable sum formula. Likewise, requiring a sigma-algebra to be closed under *arbitrary* unions would force it to be $\mathcal{P}(\mathbb{R})$ if all singletons were measurable.

Key point. A sigma-algebra specifies an admissible domain for measuring sets; it does not itself assign numerical measures. Closure of the domain and additivity of a measure are different conditions.

2. Generated Sigma-Algebras

2.1 Motivation and definition

Frequently only a collection $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ of basic events is given. It need not be a sigma-algebra, so we seek the *smallest* sigma-algebra containing these events and all their logically required countable combinations:

$$\boxed{\sigma(\mathcal{C}) := \text{the smallest sigma-algebra on } \Omega \text{ containing } \mathcal{C}.}$$

This provides the minimal *family closed under the sigma-algebra operations*, not necessarily the entire power set.

2.2 Why does the smallest one exist?

Let $\mathfrak{S} = \{\mathcal{F} : \mathcal{F} \text{ is a sigma-algebra on } \Omega,\ \mathcal{C} \subseteq \mathcal{F}\}$. It is nonempty because $\mathcal{P}(\Omega) \in \mathfrak{S}$. Define

$$\boxed{\sigma(\mathcal{C}) = \bigcap_{\mathcal{F} \in \mathfrak{S}} \mathcal{F}.}$$

An arbitrary intersection of sigma-algebras is again a sigma-algebra: Ω lies in each one, and membership in *all* of them is preserved by complements and countable unions. This intersection contains $\mathcal{C}$ and lies inside every sigma-algebra containing $\mathcal{C}$, proving minimality.

For $A \subseteq \Omega$, $\sigma(\{A\}) = \{\varnothing, A, A^c, \Omega\}$. For $\Omega = \{1,2,3,4\}$, $A = \{1,2\}$ and $B = \{2,3\}$, $\sigma(\{A,B\}) = \mathcal{P}(\Omega)$: the intersections and differences isolate all four singletons, whose finite unions generate every subset.

Key point. Generation is a minimal-closure construction: start with elementary sets and adjoin exactly what the sigma-algebra axioms force.

3. Borel Sigma-Algebras

3.1 From topology to measurability

A topology $\mathcal{T}$ contains $\varnothing, X$, is closed under arbitrary unions, and under finite intersections. It generally fails to be closed under complements: in $\mathbb{R}$, $(0,1)$ is open but its complement is not. Thus a topology need not be a sigma-algebra. The natural measurable structure associated with a topological space $(X, \mathcal{T})$ is

$$\boxed{\mathcal{B}(X) := \sigma(\mathcal{T}),}$$

the *Borel sigma-algebra*. It is the smallest sigma-algebra containing all open sets. In the usual topology of $\mathbb{R}$,

$$\boxed{\mathcal{T}_{\text{usual}} \subsetneq \mathcal{B}(\mathbb{R}) \subsetneq \mathcal{P}(\mathbb{R}).}$$

The first inclusion is strict because closed non-open sets (e.g. $[0,1]$) are Borel; the second is strict because non-Borel subsets exist.

3.2 Density and rational neighborhoods

For a topological space $(X, \mathcal{T})$, a subset $D \subseteq X$ is *dense* in X if

$$\boxed{\overline{D} = X \quad \Longleftrightarrow \quad D \cap U \neq \varnothing \text{ for every nonempty open } U \subseteq X.}$$

In particular, $\mathbb{Q}$ is dense in $\mathbb{R}$. To see this, take any $a < b$. By the *Archimedean property*, choose $n \in \mathbb{N}$ so that $n(b-a) > 1$. Setting $k = \lfloor na \rfloor + 1$ gives

$$na < k \leq na + 1 < nb, \qquad \text{hence} \qquad a < \frac{k}{n} < b.$$

Thus every nonempty open interval contains a rational number. More strongly, for any $x \in \mathbb{R}$ and open neighborhood U of x, choose $a < x < b$ with $(a,b) \subseteq U$. Rational density provides $p \in \mathbb{Q} \cap (a,x)$ and $q \in \mathbb{Q} \cap (x,b)$, so

$$\boxed{x \in (p,q) \subseteq U, \qquad p, q \in \mathbb{Q}.}$$

Hence intervals with rational endpoints form a *countable base* for the usual topology, motivating the countable Borel generator below.

3.3 Examples and countable generation

All open and closed subsets of $\mathbb{R}$ are Borel. Singletons $\{x\}$ are closed, so every countable set is Borel by countable union; in particular $\mathbb{Q}$ and, by complementation, $\mathbb{R} \setminus \mathbb{Q}$ are Borel. Intervals such as $[a, b)$ are Borel as intersections of suitable closed and open sets.

A remarkably small family generates all Borel sets:

$$\boxed{\mathcal{B}(\mathbb{R}) = \sigma\big(\{(p, q) : p, q \in \mathbb{Q},\ p < q\}\big).}$$

Reason. By the rational-neighborhood property proved above, every open $U \subseteq \mathbb{R}$ is a union of rational-endpoint intervals:

$$U = \bigcup_{\substack{p,q\in\mathbb{Q},\ p<q \\ (p,q)\subseteq U}} (p, q).$$

Since $\mathbb{Q} \times \mathbb{Q}$ is countable, this is a *countable* union, so U belongs to the generated sigma-algebra. Conversely, every generating interval is open and hence Borel.

Key point. Borel measurability is generated from topology. Countably many rational intervals already encode every open set in $\mathbb{R}$ through countable unions.

4. Measurable Spaces and Information

A *measurable space* is a pair

$$\boxed{(\Omega, \mathcal{F}), \qquad \mathcal{F} \text{ a sigma-algebra on } \Omega.}$$

No numerical size has yet been assigned. A *measure space* additionally specifies $\mu : \mathcal{F} \to [0, \infty]$ satisfying the measure axioms; probability is the special case $\mathbb{P}(\Omega) = 1$.

A sigma-algebra also describes *measurable information*. For example, let $\Omega = \{1, 2, 3, 4\}$ and consider

$$\mathcal{F} = \{\varnothing, \{1, 2\}, \{3, 4\}, \Omega\}.$$

This distinguishes which block occurred but cannot distinguish outcomes 1 from 2. If $\mathcal{F}_1 \subseteq \mathcal{F}_2$, then $\mathcal{F}_2$ is *finer*: it allows at least as many measurable distinctions.

For an observation $X : \Omega \to \mathbb{R}$, define the information it generates by

$$\boxed{\sigma(X) := \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}.}$$

The inverse-image identities guarantee that this is a sigma-algebra. If $X(1) = X(2) = 0$ and $X(3) = X(4) = 1$, then $\sigma(X) = \{\varnothing, \{1, 2\}, \{3, 4\}, \Omega\}$. This is the prototype for information carried by random variables and filtrations.

5. Topology, Sigma-Algebras, and Maps

Both topologies and sigma-algebras are families of subsets stable under specified operations, but their closure requirements differ:

- **Topology:** arbitrary unions and finite intersections; complements need not be open.
- **Sigma-algebra:** complements and countable unions (hence countable intersections); arbitrary unions need not be measurable.

They support parallel notions of structure-preserving maps. For topological spaces $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$,

$$\boxed{f : X \to Y \text{ continuous} \iff f^{-1}(U) \in \mathcal{T}_X \quad \forall U \in \mathcal{T}_Y.}$$

For measurable spaces $(X, \mathcal{F})$, $(Y, \mathcal{G})$,

$$\boxed{f : X \to Y \text{ measurable} \iff f^{-1}(B) \in \mathcal{F} \quad \forall B \in \mathcal{G}.}$$

Both definitions use *preimages* because they preserve complements and arbitrary unions/intersections. In particular, every continuous map between topological spaces is Borel measurable when both spaces carry their Borel sigma-algebras: the preimages of target open sets are source open sets, and the same conclusion extends to their generated sigma-algebras.

Key point. Topology encodes local closeness; a sigma-algebra encodes measurable events. Preimages provide the common language for maps preserving either structure.